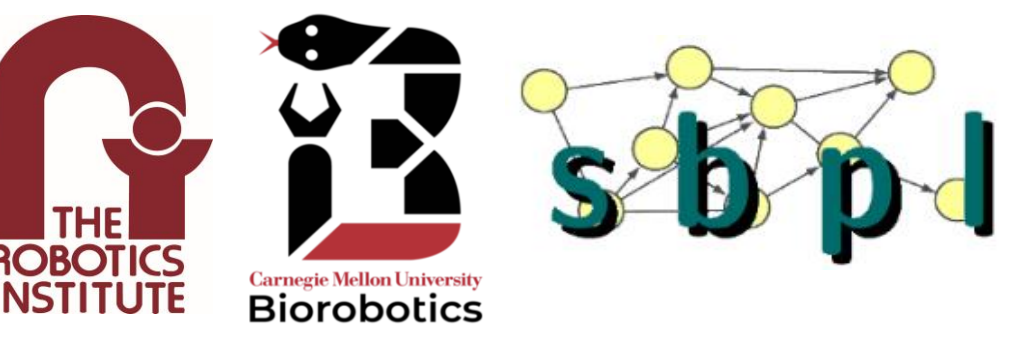


Think Fast: Real-Time Kinodynamic Belief-Space Planning for Projectile Interception



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Motivation

Problem: Intercepting fast-moving objects with robots requires decisions within very tight time constraints [1,2]

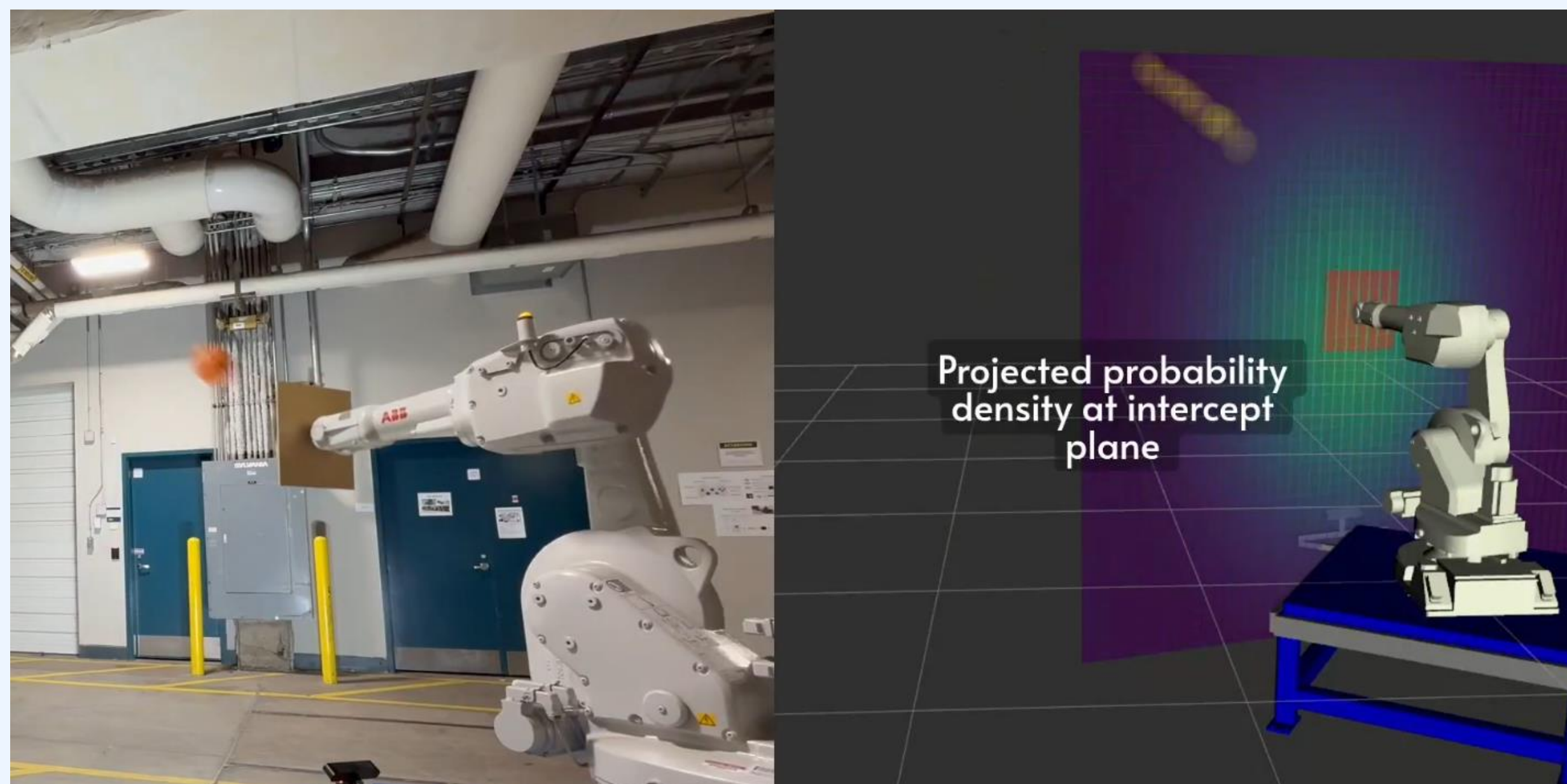
Uncertainty from sensors: Noisy, incomplete sensor data leads to uncertainty in predicting the target's future state

Planning under uncertainty: The robot must begin moving toward interception while still receiving sensor data and updating its predictions

Approach

Closed loop planning and execution framework compatible with kinodynamic capabilities of robot manipulators:

- **Offline:** build action tree from minimum-time, jerk-limited motion primitives, encoding reachability to many goals from the home configuration.
- **Online:** update belief over goal states at sensor rate (50 Hz) and re-evaluate a QMDP value function over the tree — enabling reactive replanning to maximize intercept probability

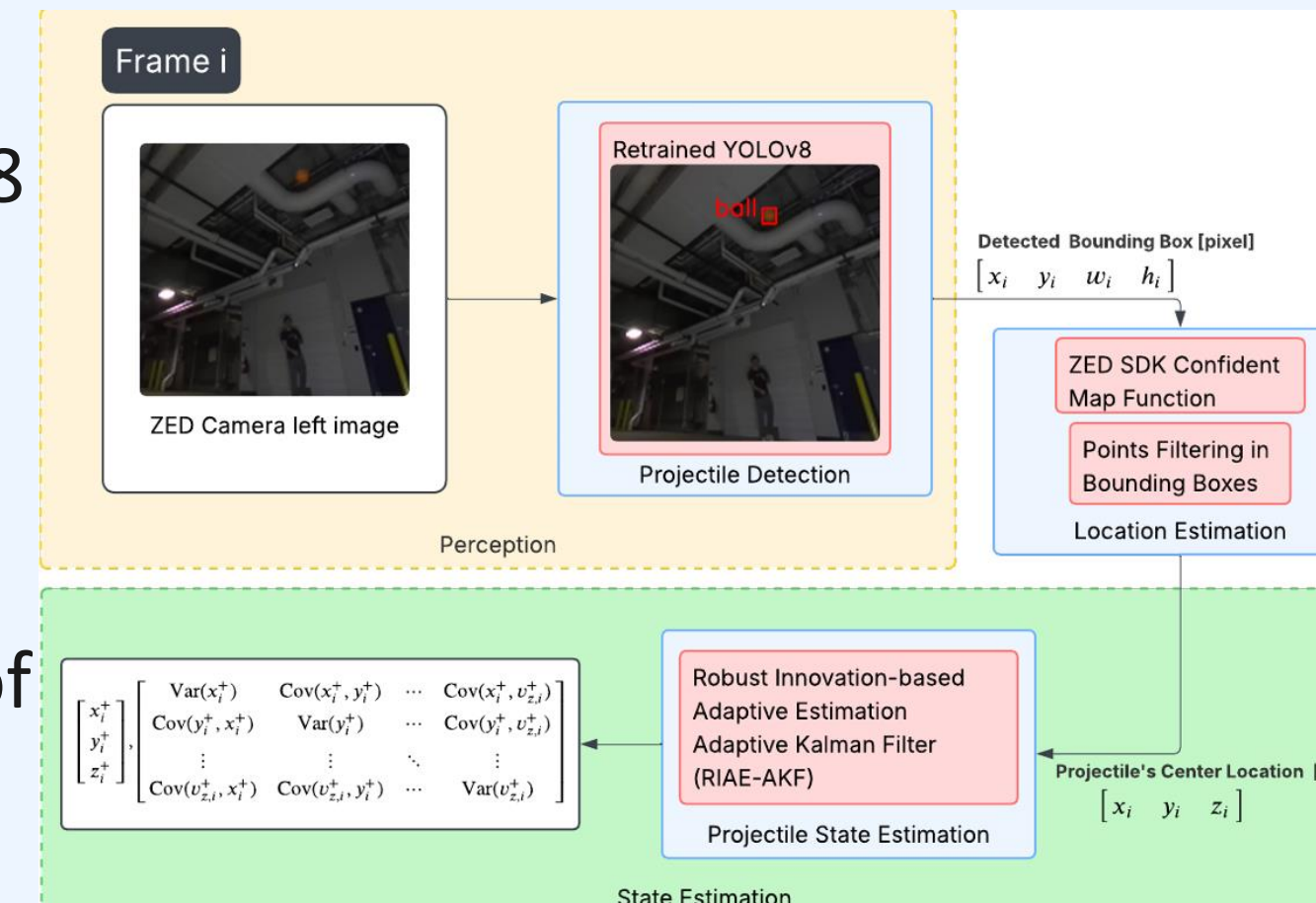


Method

State Estimation

Target Tracking

- Segment the projectile from the RGB-D camera frames with a fine-tuned YOLOv8 detector [3]
- Estimate state under projectile motion with adaptive Kalman filter (RIAE-AKF) - accounts for distant-dependent noise[4]
- Yields a 6D state (μ) and covariance (Σ) of the projectile (x, y, z, v_x, v_y, v_z) for downstream planning



Belief Over Goal States in Robot Frame

1) Crossing time at the intercept plane $x = x^*$:

$$\tau = \frac{x^* - X}{V_x}$$

with

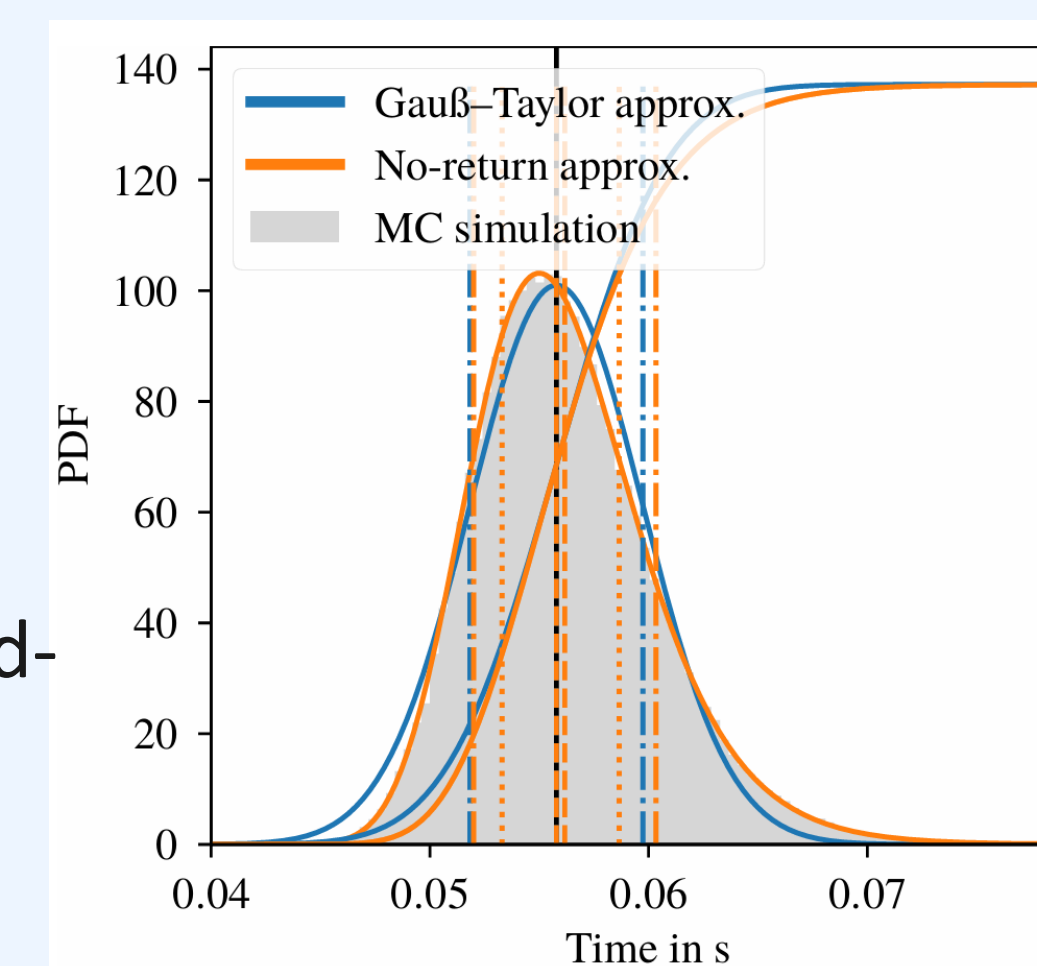
$$\mathbb{E}[\tau] \approx \bar{\tau} = \frac{x^* - \mu_X}{\mu_{V_x}}, \quad \text{Var}[\tau] \approx g_\tau \Sigma g_\tau^\top$$

2) First-order linearization about μ gives a closed-form Gaussian over interception state [5]:

$$o \sim \mathcal{N}(h(\mu), J \Sigma J^\top)$$

where

$$o = (Y_{\text{plane}}, Z_{\text{plane}}, \tau)$$



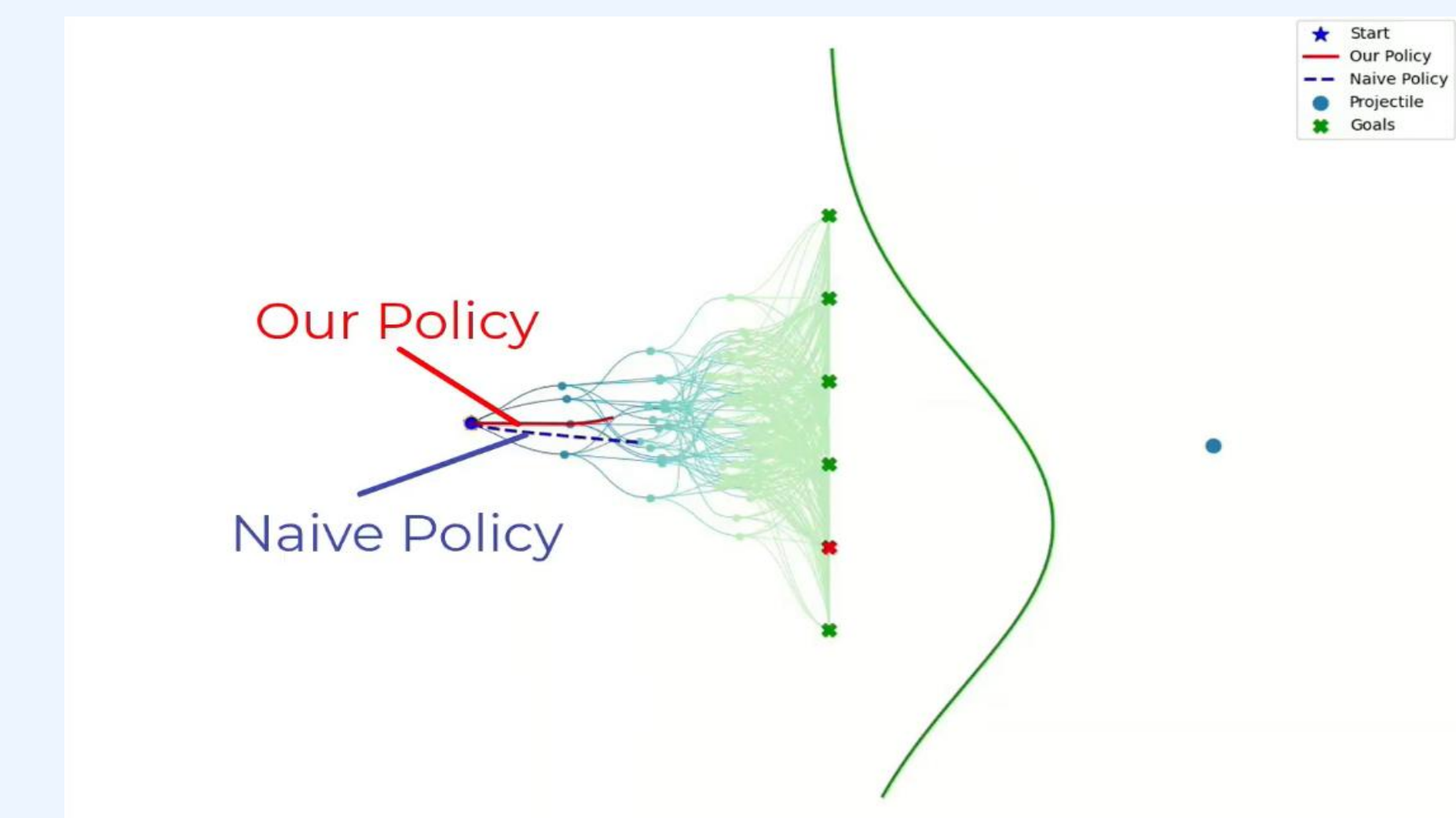
Planning

An action tree rooted at the home configuration encodes reachability through state-time space. Each node stores (joint position, velocity, acceleration) and time-to-come from the root.

- Action Tree: expand iteratively with minimum-time motion primitives toward sub-goals discretizing the front plane of the manipulator's workspace
- Minimum-time Optimal Control: jerk-limited primitives [6] satisfy both kinematic and dynamic constraints, maximizing aggressiveness while limiting motor strain
- Value Function: QMDP approximation of expected intercept success – cheap upper bound on optimal value of full POMDP – assumes fully observable state after one MDP transition [7]

$$V(n, b(o)) = \mathbb{E}_{o \sim b} [R(n, o) + \max_{a \in \mathcal{A}} V(n', b(o))]$$

- Online value backups as the belief evolves $\rightarrow$ greedy next action



Experiments & Results

Integrated system tested on IRB-1600 6DOF manipulator + ZED 2i stereo camera. Balls thrown from 6–8 m; time of flight 400–800 ms.

- Our method: **74% blocking success** across 50 throws.
- Baseline (Deterministic Planning): 63% blocking success on the same setup.
- Encoding time-constrained reachability makes our planner robust to evolving observations; baseline planner commits immediately and struggles under noisy early estimates.



References

- [1] B. Bäuml et al. "Kinematically optimal catching a flying ball with a hand-arm-system."
- [2] R. Natarajan et al. "Preprocessing-based kinodynamic motion planning for intercepting projectiles."
- [3] G. Jocher, A. Chaurasia, J. Qiu. "Ultralytics YOLOv8."
- [4] X. ZhiWen et al. "Robust innovation-based adaptive Kalman filter for INS/GPS land navigation."
- [5] M. Reith-Braun, F. Pfaff, J. Thumm, and U. D. Hanebeck, "Approximate first-passage time distributions for gaussian motion and transportation models."
- [6] L. Berscheid, T. Kröger. "Jerk-limited real-time trajectory generation with arbitrary target states."
- [7] M. Littman, A. Cassandra, L. Kaelbling. "Learning policies for partially observable environments: Scaling up"